

Keiko Woo Wilkins

University of Hawai'i at Mānoa
Kewalo Marine Laboratory

keikow2@hawaii.edu
www.keikowilkins.com

Education

Doctor of Philosophy, Marine Biology **December 2025**

University of Hawai'i at Mānoa (Honolulu, HI)

Dissertation Research: *Tiny Particles, Big Problems: Microplastic Lessons from Coral Reproduction to Reef Conservation*

Master of Science, Biology **January 2020**

Miami University (Oxford, OH)

Certificate: Applied Statistics

Thesis Research: *The Importance of Dissolved Organic Matter Source on the Survival and Growth of Juvenile Daphnia*

Bachelor of Science, Biological Sciences **January 2018**

Miami University (Oxford, OH)

Minor: Chinese

Senior Capstone project: *The Effects of Hydrogen Sulfide within the Hypolimnion of a Dystrophic and Eutrophic Lake on Zooplankton Survival and Vertical Distribution*

Research Experience

NOAA Dr. Nancy Foster Scholar **July 2022 – Present**

National Oceanic and Atmospheric Administration

Principal Investigator: Dr. Robert Richmond

- Selected for the prestigious Dr. Nancy Foster Scholarship to explore *The Physical and Chemical Analysis of Coral Ingested Microplastics Across the Pacific*
- Coordinated coral and water sample collections with NOAA National Marine Sanctuaries of the Pacific Region including Hawaiian Islands Humpback Whale National Marine Sanctuary, National Marine Sanctuary of American Samoa, Rose Atoll National Wildlife Refuge and Papahānaumokuākea Marine National Monument
- Conducted education and outreach activities in American Samoa in collaboration with National Marine Sanctuary of American Samoa to create teaching materials and facilitate laboratory based experiential learning activities with 28% of 1-12th grade schools in American Samoa

Graduate Research Assistant **August 2020 – July 2022**

Kewalo Marine Laboratory

University of Hawai'i at Mānoa (Honolulu, HI)

Principal Investigator: Dr. Robert Richmond

- Responsible for collecting data for coral ecotoxicological exposures including protein extraction, quantification, and digestion
- Collaboration with U.S. Fish and Wildlife Service to explore *The Effects of Tungsten on Hawaiian Coral Health, Reproduction and Larval Development*
- Mentored 14 undergraduate students to allow them to explore coral assessment techniques and experimental design

Limnology and Oceanography Research Exchange

January 2020 – March 2020

Association for the Sciences of Limnology and Oceanography

Southern Cross University (Lismore, Australia)

Principal Investigator: Dr. Elisabeth Deschaseaux

- Accepted into the first cohort of ASLO LOREX students to participate in the National Science Foundation-funded international research collaboration program
- Responsible for collecting light, temperature, photosynthetic efficiency, and chlorophyll-a data from *Duncanopsammia axifuga* during the two-week experiment
- Collaboration with Dr. Deschaseaux to explore *The Effects of Elevated Temperature and High Dimethylsulfoniopropionate on Coral Reef Feeding Rates*

Graduate Research Assistant

January 2018 – January 2020

Global Change Limnology Laboratory

Miami University (Oxford, OH)

Principal Investigator: Dr. Craig Williamson

- Responsible for summer mid-month field sampling of chemical, biological, and physical data of several lakes in the Poconos, PA
- Mentored two undergraduate students, one of whom was accepted into the Louis Stokes Alliance for Minority Participation Summer Program
- Completed *The Effects of Hydrogen Sulfide within the Hypolimnion of a Dystrophic and Eutrophic Lake on Zooplankton Survival and Vertical Distribution*

Education and Outreach Initiatives

Columbus School for Girls

March 2025

- Reach: 10 high school juniors and seniors (Bexley, Ohio)
- Hour-long AP Environmental Sciences class lecture on microplastic pollution and endocrine disruption in the marine environment

American Samoa Educator Workshop

February 2025

National Marine Sanctuary of American Samoa (Utulei, American Samoa)

- Reach: 5 educators at elementary and high schools in American Samoa
- Hour-long session on the importance of coral reef ecosystems, key terms for students to know and global vs. local stressors affecting coral reef ecosystems

American Samoa 1st-12th Grade Schools

February 2025

National Marine Sanctuary of American Samoa (Utulei, American Samoa)

- Reach: 300 elementary – high school students on Tutuila
- 11, hour long hands-on activities on coral bleaching and ocean acidification effects on coral reefs along with plastic pollution research happening in American Samoa

Darwin Day

February 2024

University of Hawai'i at Mānoa (Honolulu, HI)

- Reach: ~50 local community members and local university students
- 4-hour long session with marine invertebrate touch tanks and information sharing

Canton City High School and Middle Schools

January 2024

Stark County Educational Service Center (Canton, Ohio)

- Reach: 549 middle and high schoolers in Canton, Ohio
- 7, hour-long sessions connecting landlocked locations to impacts on the ocean, discussion of plastic pollution in Ohio and the ocean, and the importance of corals and the ocean to Ohio

Canton City Community

January 2024

Stark County Educational Service Center (Canton, Ohio)

- Reach: 68 local community members including college professors, students, and families
- Hour-long session connecting pollution in Ohio to pollution in the ocean, microplastic impacts on humans and marine life, the importance of coral reef ecosystems, and how corals impact humans

Ocean Guardians Youth Ambassadors

November 2023

National Oceanic and Atmospheric Administration

- Reach: ~ 30 middle, high school, and college students across the US
- 30-minute session sharing my journey to marine biology and my current research with NOAA

American Samoa Community College

November 2023

American Samoa Community College (Pago Pago, American Samoa)

- Reach: 20 community college students and community members in American Samoa
- Hour-long session sharing my journey to marine biology and opportunities for local students to be involved in research opportunities in Hawaii

American Samoa 1st-12th Grade Schools

November 2023

National Marine Sanctuary of American Samoa (Utulei, American Samoa)

- Reach: 300 students representing 28% of 1st-12th grade schools on Aunu'u and Tutuila, American Samoa
- 11, hour-long experiential learning workshops on the importance of coral reef ecosystems, threats to corals include microplastic pollution with activities including water quality assessments and analysis of beach samples

93KHJ Radio Interview**November 2023**

National Marine Sanctuary of American Samoa (Pago Pago, American Samoa)

- Reach: most listened-to radio station in American Samoa
- 30-minute interview sharing microplastic pollution impacts on coral reefs and dissertation research in American Samoa

School of Ocean & Earth Science & Technology Open House**October 2023**

University of Hawai'i at Mānoa (Honolulu, HI)

- Reach: ~100 local youth and families from Oahu, HI
- 4-hour long with marine invertebrate touch tanks and information-sharing

Youth Explorer's Program**June 2023**

University of the Virgin Islands (St. Croix, USVI)

- Reach: 40 middle-high-school students in the Virgin Islands
- Hour-long interactive session on marine debris and its impacts on our planet

Marine Biology Club**March 2023**

Miami University (Oxford Ohio)

- Reach: 20 undergraduate students at Miami University
- Hour-long session sharing my journey to marine biology and Miami University initiatives which made it possible

Teaching Experience***Teaching Assistant*****Spring 2022**

OCN 105/SUST 115: Sustainability in a Changing World

Department of Oceanography

University of Hawai'i at Mānoa (Honolulu, HI)

- Undergraduate level – 1 credit hour
- Responsible for managing 400 students including helping with course design, grading of free response assignments, sharing of course content, hosting weekly review sessions, creation of quiz and exam questions

Teaching Assistant**Summer 2021**

BIO 101/ SUST 111 Laboratory: Biology and Sustainability in Society

Department of Biology

University of Hawai'i at Mānoa (Honolulu, HI)

- Undergraduate level – 1 credit hour
- Responsible for hosting online laboratory sessions and helping students in the completion of student independent research projects

Teaching Assistant**Fall 2018**

BIO/MBI 115 Laboratory: Biological Concepts: Ecology, Evolution, Genetics, and Diversity
 Department of Biology
 Miami University (Oxford, OH)

- Undergraduate level - 1 credit hour
- Department of Biology Graduate Teaching Evaluation Instructor Rating: 3.8/4.0

Teaching Assistant**Spring 2018**

BIO/MBI 116 Laboratory: Biological Concepts: Structure, Function, Cellular, and Molecular Biology
 Department of Biology
 Miami University (Oxford, OH)

- Undergraduate level - 1 credit hour
- Department of Biology Graduate Teaching Evaluation Instructor Rating: 3.6/4.0

Training and Certifications***Open Science Certification*****March 2023**

Pathways to Open Science

- In-progress course on conducting open science through data science literacy and community building

Scientific Diver**June 2022**

American Academy of Underwater Sciences

- Successful completion of over 100 hours of written practice and in-water skills evaluations

Teaching English to Speakers of Other Languages**August 2019**

International Open Academy

- Attended 120-hour TESOL Certification course to gain a better understanding of the basics of linguistics and develop effective teaching strategies

Graduate Student Teaching Enhancement Program Core Member**May 2019**

Center for Teaching Excellence
 Miami University (Oxford, OH)

- Assessed current teaching strategies and ways to improve teaching in the future through workshops and group discussions with faculty and graduate students

Mental Health First Aid Certification**January 2019**

National Council for Behavioral Health

Miami University (Oxford, OH)

- Attended 10-hour certification workshop to further my knowledge of mental health crises that may be affecting students and tools to help students in need

Pedagogy

August 2018

Department of Biology

Miami University (Oxford, OH)

- Learned teaching strategies to become an effective biology teaching assistant in the biology department

Honors and Awards

Ocean Odyssey Marine Debris Award for DEIJA

2024 – Present

National Marine Sanctuary Foundation and National Oceanic and Atmospheric Administration Marine Debris Program

- Awarded funding for travel to American Samoa for student and educator workshop on isolating marine debris from environmental samples, hands-on activities about coral threats and supplies for sample processing
- Award amount: \$7500

Tidal Wave Fellowship

November 2023

Black in Marine Science

- Awarded funding for travel and participation in both the 2023 Black in Marine Science Wellness Retreat (Norfolk, VI) and the 2024 United Nations Ocean Decade Conference (Barcelona, Spain)
- Award amount: airfare, housing, and registration for both conferences

Garrett Lynch Memorial Scholarship

January 2023

University of Hawai'i at Mānoa (Honolulu, HI)

- Awarded funding for research supplies to cover the cost of travel to American Samoa for coral collections
- Award amount: \$2000

The Charles H. and Margaret B. Edmondson Research Fund

January 2023

University of Hawai'i at Mānoa (Honolulu, HI)

- Awarded funding for research supplies to cover the cost of travel to American Samoa for coral collections
- Award amount: \$2000

18 of the last 9

January 2023

Miami University (Oxford, OH)

- Awarded Miami University's version of Forbes 30 under 30 as one of 18 alumni who graduated within the last 9 years for leadership, creativity, and significant positive impact on the world

Multicultural Program

December 2022

Association for the Sciences of Limnology and Oceanography

- Awarded funding to participate in Ocean Sciences Meeting in Mallorca, Spain
- Award amount: airfare, housing, abstract fees and registration fees

Dr. Nancy Foster Scholarship

June 2022

National Oceanic and Atmospheric Administration (Washington, DC)

- Selected for highly competitive NOAA program to provide funding for the remainder of PhD studies and professional development opportunities

Multicultural Program

December 2021

Association for the Sciences of Limnology and Oceanography

- Awarded funding to participate in Ocean Sciences Meeting virtually
- Award amount: registration and abstract fees

Career and Leadership Development Grant

November 2021

American Advancement of University Women Honolulu Chapter (Honolulu, HI)

- Awarded grant for travel and registration support to attend and present research at the Joint Aquatic Sciences Meeting 2022 in Grand Rapids, MI

Graduate Student Organization Award

November 2021

University of Hawai'i at Mānoa (Honolulu, HI)

- Awarded grant for research funds to travel to Maui and collect coral samples for microplastic ingestion study

Diversity Enhancement Pathway Graduate Assistantship

January 2019 – January 2020

Biology Research Assistant

Miami University Graduate School (Oxford, OH)

- Responsible for research within the global change limnology laboratory

Graduate Student Teaching Enhancement Program Grant

Spring 2019

Center for Teaching Excellence

Miami University (Oxford, OH)

- Awarded for successful completion of semester long workshops focused on incorporating new teaching strategies into the classroom including new classroom assessment

techniques, tools for respectful dialogues and incorporating research experiences into the classroom

Publications

- Wilkins, K.W.** (2025). *Tiny Particles, Big Problems: Microplastic Lessons from Coral Reproduction to Reef Conservation* (Doctoral dissertation). University of Hawai'i at Mānoa (in review)
- Wilkins, K. W., & Richmond, R. H.** (2025). Unseen threats: negative effects of microplastic leachate on coral planulae settlement. *Frontiers in Marine Science*, 12, 1596594.
- Wilkins, K. W.** (2024). Living in a Plastic Ocean: Addressing Plastic Pollution Threats to Corals in the Pacific. *Asia-Pacific Journal of Ocean Law and Policy*, 9(2), 187-197.
- Wilkins, K. W., Yew, J. Y., Seeley, M., & Richmond, R. H.** (2024). Microplastic leachate negatively affects fertilization in the coral *Montipora capitata*. *Integrative and Comparative Biology*, 64(4), 1131-1140.
- Isma, L.M., M.L. Bernard, J. Layton, R. Santana, **K.W. Wilkins**, and D. Nembhard. (2023). Black women in ocean science: Barriers to advancement and strategies to move forward. *Oceanography* 36(4), <https://doi.org/10.5670/oceanog.2024.129>.
- Wilkins, K. W., & Rosa Marín, A.** (2021). Creating More Inclusive Waters in the Aquatic Sciences. *Limnology and Oceanography Bulletin*.
- Hughes, H., Cziesielski, M. J., Love, C., Nowakowski, C., Marín, A. R., & **Wilkins, K. W.** (2021). Looking to the Future: Perspectives from the Next Generation of Coral Scientists. *Limnology and Oceanography Bulletin*.
- Love, C., Cziesielski, M. J., Hughes, H., Nowakowski, C., Rosa Marín, A., & **Wilkins, K. W.** (2021). Rethinking Collaboration for Coral Reef Science. *Limnology and Oceanography Bulletin*.
- Wilkins, K. W., Rosa-Marín, A., Cziesielski, M., Hughes, H., Love, C., & Nowakowski, C.** (2021). Short-and Long-Term Visions for Protecting Coral Reefs. *Limnology and Oceanography Bulletin*.
- Cziesielski, M. J., Hughes, H., Love, C., Marín, A. R., Nowakowski, C., **Wilkins, K.** Channeling Hope for Reefs: A Series of Perspectives from Young Coral Reef Scientists. (2021). *Limnology and Oceanography Bulletin*.
- Nowakowski, C., Hughes, H., Love, C., Marín, A. R., **Wilkins, K., & Cziesielski, M. J.** Communicating Hope for Coral Reefs. (2021). *Limnology and Oceanography Bulletin*.
- Wilkins, K. W.** and Burnett, S. (2021). Student Mentoring Programs in Aquatic Science Societies. *North American Lake Management Society LakeLine*, 40(4), 35-37.
- Rosa-Marín, A., & **Wilkins, K. W.** (2020). Challenging traditional international research collaborations. *Limnology and Oceanography Bulletin*, 29(3), 76-78.
- Wilkins, K. W., Rosa-Marín, A., Weisend, R. E., & Kurosawa, E.** (2020). Benefits, Challenges, and Opportunities for International Research Collaborations: Perspectives from the Limnology and Oceanography Research Exchange. *Limnology and Oceanography Bulletin*, 29(4), 125-130.
- Williamson, C. E., Overholt, E. P., Pilla, R. M., & **Wilkins, K. W.** (2020). Habitat-mediated responses of zooplankton to decreasing light in two temperate lakes undergoing long-term browning. *Frontiers in Environmental Science*, 8, 73.

- Wilkins, K. W.,** Overholt, E., & Williamson, C. (2020). The effects of dissolved organic matter from a native and an invasive plant species on juvenile *Daphnia* survival and growth. *Journal of Plankton Research*.
- Wilkins, K.W.** (2020). *The Importance of Dissolved Organic Matter Source on the Survival and Growth of Juvenile Daphnia* (Master's thesis). Available from ProQuest Dissertations & Theses Global database. (UMI No. 27818670)
- Wilkins, K. W.** (2020). Different Types of Tea. *North American Lake Management Society LakeLine*, 40(1), 17-19.
- Beck, H. N., Cohen, A., McKenzie, T., Weisend, R., **Wilkins, K. W.**, & Woodstock, M. S. (2019). Broadening Horizons: Graduate Students Participating in International Collaborations through the Limnology and Oceanography Research Exchange (LOREX) Program. *Limnology and Oceanography Bulletin*, 28(3), 85-89.

Professional Affiliations & Community Involvement

Mālama Pūpūkea-Waimea

Visitor Impact Surveyor

November 2023 – January 2024

Black in Marine Science

Member

November 2020 – Present

Society of Black Trainees

Member

December 2020 – 2024

Consortium of Aquatic Science Societies

Diversity, Equity and Inclusion Ambassador

November 2020 – 2024

American Geophysical Union

Member

September 2020 – Present

Black Women in Ecology, Evolution and Marine Science

Community Coordinator

March 2025 – Present

Membership Management and Program Coordinator

February 2022 – March 2025

Mentor

March 2021– Present

Member

August 2020 – Present

North American Lake Management Society

Justice, Equity, Diversity, and Inclusion program Co-chair

January 2021 – December 2024

Diversity Task Force member

June 2020 – January 2021

Member

April 2019 – Present

Association for the Sciences of Limnology and Oceanography

Limnology and Oceanography Research Exchange Evaluator

June 2020 – December 2020

Member

November 2018 – Present